Assignment # 2 :-

Question # 1 :-

The OSI Model is a conceptual framework that explans how data is sent and recieved over a network. It is divided into seven sepecific layers, each performing a unique function.

Application layer (layer 7)

* **Function:** This is the topmost that interacts directly with user applications, providing network services like email or file transfers.
* **Example:** A web browser acessing a website using HTTP.

Present ation layer (layer 6)

* **Function:** This layer translates, encrypts and compresses data so that it is formed properly for the recieving system.
* **Example:** Encrypting a password before it is sent over the internet or formatting an image as a JPEG.

Session layer (layer 5)

★ Function: It establishes, maintains and securely terminates connection (or "sessions") between two interacting devices.

★ Example: A live video conferecing call that stays connected until you hang up.

Transport layer (layer 4)

★ Function: This layer ensures that the entire message is delivered reliably from end to end breaking large data into smaller segments.

★ Example: Sending a large email attachment where TCP ensures no part of the file are lost during transfer.

Network layer (layer 3)

★ Function: This layer organizes data into frames logical addressing and routing, finding the best physical path for data to travel from one network to another.

★ Example: A home router using IP addresses to send data to the correct computer.

Data link layer (layer 2)

* **Function:** This layer organizes data into frames, handles physical addressing and checks for physical errors between directly connected nodes.
* **Example:** A network switch using MAC addresses to deliver data to a specific laptop on a local network.

Physical layer (layer 1)

* **Function:** The lowest layer is responsible for sending raw data bits (0s and 1s) over physical hardware media.
* **Example:** Copper ethernet cables, fiber optics cables or WiFi radio waves.

How data moves through these layers:

When a user sends data, it travels from the application layer down to the physical layer in the process called encapsulation. At each step down, the layer adds its own control information (called a header) to the data. Once the data travels across the physical networks to the destination device, it moves back up from the physical layer to the

Application layer. This reverse process is called decapsulation, where each layer reads and remove its specific header until the original message is successfully delivered to the recieving application

Question no 2 :-

The TCP / IP Protocol Suite is the foundational set of rules that governs internet communication. Unlike the OSI model, it groups networking functions into a streamlined four-layer structure

Application layer

★ **Function :** This layer combines the top three layers of the OSI layer model. It is responsible for allowing applications to access the network and provides user interfaces and communication services.

★ **Common Protocol :** HTTP (Hypertext Transfer protocol) which is used to securely fetch and display web pages on the internet.

Transport layer

★ **Function:** This layer provides host-to-host communication and ensures that data is delivered reliably It tracks data segments and manages the speed and sequence of the transfer

★ Common Protocol : TCP (Transmission Control Protocol) which guarantees that data is delivered accurately and in the correct order by requesting retransmission of any last packets.

Internet layer

★ Function: This layer focuses on addressing and routing data packets across independent networks to ensure they reach their final destination

★ Common Protocol: IP (Internet Protocol), which assigns logical addresses (IP addresses) to devices and routes the data packets based on those student addresses

Network Access layer

★ Function: The lowest layer deals with the physical delivery of data It manages how a computer connects to the

★ Common Protocol : ARP (Address Resolution Protocol) which is used to map a known logical

IP address to a physical hardware MAC address so the data can be delivered locally.

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